$L^a = J^{ab}\omega_b$
$J^{ii} = \int x^{j^2} - x^{k^2} dm(j, k \neq i)\ dm = pdV = \frac{m}{a^3}dxdydz$
$J^{11} = \int_V y^2 + z^2 dm \int y^2 + z^2 pdxdydz = ay^2 + az^2 dydz = \frac{1}{3}a * a^3 + a^2 z^2 dz = \frac{1}{3}a^2 * a^3 + \frac{1}{3}a^2 a^3 = \frac{2}{3}a^5$
$J^{11} = \frac{m}{a^3}\frac{2}{3}a^5 = \frac{2}{3}ma^2$
$J^{ii} = \frac{2}{3}ma^2$
$J^{12} = -\int xypdxdydz = -\frac{1}{2}a^2 pydydyz = -\frac{1}{4}a^4 pdz = -\frac{1}{4}a^5 p = -\frac{1}{4}ma^2$
$J^{\nu\mu} = -\frac{1}{4}ma^2$
$L^a = J^a_b w^b; w^b = \frac{1}{\sqrt{3}}(i^b + j^b + k^b)$
$L^a = \frac{1}{\sqrt{3}}(J^{a1} + J^{a2} + J^{a3}) = \frac{1}{\sqrt{3}}((\frac{2}{3}ma^2 - \frac{1}{2}ma^2)i^a + (\frac{2}{3}ma^2 - \frac{1}{2}ma^2)j^a + (\frac{2}{3}ma^2 - \frac{1}{2}ma^2)k^a)$
$= \frac{1}{\sqrt{3}}(\frac{1}{6}(j^a + j^a + k^a))ma^2$
$T = \frac{1}{2}w^a J_{ab} w^b = \frac{1}{2}\frac{1}{\sqrt{3}}(i^b + j^b + k^b)\frac{1}{\sqrt{3}}(\frac{1}{6}(j^a + j^a + k^a))ma^2 = \frac{1}{12}ma^2$
answer(1)

$$\frac{1}{\sqrt{3}}(\frac{1}{6}(j^a + j^a + k^a))ma^2 \tag{1}$$

answer(2)

$$T = \frac{1}{12}ma^2 \tag{2}$$

$m\frac{dv}{dt} = -\frac{1}{2}v\ m\frac{dlnv}{dt} = -\frac{1}{2}\ lnv = C - \frac{t}{2m}\ ;\ v = Ce^{-\frac{t}{2m}}\ ;\frac{1}{2}mv^2 = mgR$ $v = \sqrt{2gR}$
$v|_{t=0} = C = \sqrt{(2gR)}\ ;\ v = \sqrt{(2gR)}e^{-\frac{t}{2m}}$

# 1 Oct 22,example

## 1.1 metter = m stick.And with a m ball.

$\frac{1}{2}\frac{1}{3}ml^2\omega^2 = \frac{1}{2}mgl;\omega = \sqrt{\frac{3g}{l}}$
$\frac{1}{3}ml^2\omega = \frac{1}{3}ml^2\omega_0 + ml^2\omega_0 4; \omega_0 = \frac{1}{4}\omega$
$2mg\frac{3}{4}l(1 - cos\theta) = \frac{1}{2}\frac{1}{3}ml^2\omega_0^2 + \frac{1}{2}ml^2\omega_0^2 = \frac{2}{3}ml^2\omega_0^2 = \frac{2}{3}ml^2\frac{3g}{16l} = \frac{1}{8}mgl$
$\frac{1}{12} = (1 - cos\theta); cos\theta = \frac{11}{12}$

## 1.2 a stick m and 60 degree.

$L = \frac{1}{3}ml^2$
$\frac{1}{2}\frac{1}{3}ml^2\omega^2 = mg\frac{1}{2}lsin30;\omega = \sqrt{\frac{3g}{2l}}$
$mgcos60 - T_1 = -m\frac{1}{2}l\omega^2; T_1 = \frac{1}{2}mg + \frac{1}{2}ml\frac{3g}{2l} = \frac{5}{4}mg$
$\frac{1}{3}ml^2\beta = \frac{1}{2}mglsin60;\beta = \frac{3\sqrt{3}g}{4l}$
$a_c = \beta\frac{1}{2}l = gsin60 + T_0/m; T_0 = \frac{\sqrt{3}}{8}mg$

$|\boldsymbol{T}| = \sqrt{(\frac{103}{64})}mg$

***way2* :** $a = a' + a_c + 2v' \times \omega + r' \times \dot{\omega} + r' \times \omega \times \omega$

$r = r' + r_c; v = v' + v_c + r' \times \omega;$ $a$=$a' + a_c + v' \times \omega + v' \times \omega + r' \times \dot{\omega} + r' \times \omega \times \omega$

$a' = a - a_c - 2v' \times \omega - r' \times \dot{\omega} - r' \times \omega \times \omega$

$r' \times a' = -r' \times a_c - r' \times r' \times \dot{\omega} = r' \times \beta' \times r' = -\frac{1}{2}l^2\beta = \frac{1}{2}l^2\frac{3\sqrt{3}g}{4l}; \frac{3\sqrt{3}gl}{8}\frac{1}{12}ml^2 = \frac{1}{4}l^2F_n; F_n = \frac{\sqrt{3}}{8}mg$

$\frac{3\sqrt{3}mgl}{8}$

$\frac{1}{12}ml^2\frac{\sqrt{3}gl}{8} = \frac{1}{2}l \times F_n$

## 1.3 Oct 27

the ball have angler velocity,ant floor have friction force:

way 1:

$I\omega_0 - I\omega = \mu mgR\Delta t$

$m\omega R = \mu mg\Delta t$

so we get $\omega = \frac{2}{7}\omega_0$

and the $\Delta t = \frac{\frac{2}{7}\omega_0 R}{\mu g}$

so x = $\frac{1}{2}\mu g t^2 = \frac{2}{49\mu g}\omega_0^2 R^2$

way2:

$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{5}mR^2\dot{\theta}^2$

$v = 0; W = \mu mgx - \mu mgR\theta$

$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{5}mR^2\dot{\theta}^2 + \mu mgx - \mu mgR\theta$

$m\ddot{x} = \mu mg; \frac{2}{5}mR^2\ddot{\theta} = -\mu mgR$

$m\Delta\dot{x} = \mu mg\Delta t; \frac{2}{5}mR^2\Delta\dot{\theta} = -\mu mgR\Delta t$

$mv = \mu mg\Delta t; \frac{2}{5}mR^2(\omega - \omega_0) = -\mu mgR\Delta t; \omega = vR$

$m\omega R = \mu mg\Delta t$ ; $\frac{7}{5}mR^2\omega = \frac{2}{5}mR^2\omega_0; \omega = \frac{2}{7}\omega_0$

$\Delta t = \frac{2\omega_0 R}{7\mu g}$

$\Delta x = \frac{1}{2}\mu g(\frac{2\omega_0 R}{7\mu g})^2 = \frac{2\omega_0^2 R^2}{49\mu g}$

## 1.4 Nov 10

3 Lagrangian questions

(A) a surface with a heavy object and a turintable.

$L = \frac{1}{2}m_1\dot{l}^2 + \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 - (C - m_2gl)$

$\frac{3}{2}m_1\ddot{l} + m_2\ddot{l} = m_2g$

$\ddot{l} = \frac{m_2g}{\frac{3}{2}m_1 + m_2}$

(B) a ramp with a turintable

$L = \frac{3}{4}m\dot{l}^2 - (C - mgsin\theta l)$

$\frac{3}{2}m\ddot{l} = mgsin\theta$

$\ddot{l} = \frac{2}{3}gsin\theta$

(C) A mass-have turntable tite with a heavy object

$L = \frac{1}{2}m'\dot{l}^2 + \frac{1}{2}\frac{1}{2}m\dot{l}^2 - (C - m'gl)$

$m'\ddot{l} + \frac{1}{2}m\ddot{l} = m'g$
$\ddot{l} = \frac{m'g}{m' + \frac{1}{2}m}$